Supplementary Material S1

Structure-Preserving Residual Learning and Predictive Control of Nonlinear Descriptor Power-Network Models

Sri Venkata Durga Sudarsan Madhyannapu

This file supports the manuscript submitted to *Communications in Nonlinear Science and Numerical Simulation*. Every number below is read from the CSV files in the `results/` folder of the repository. Section 0 holds figures and tables moved out of the manuscript.

0 Material moved from the main text

This section holds the figures and tables that were moved out of the manuscript to keep it short. The manuscript refers to this material as Supplementary Material S1.

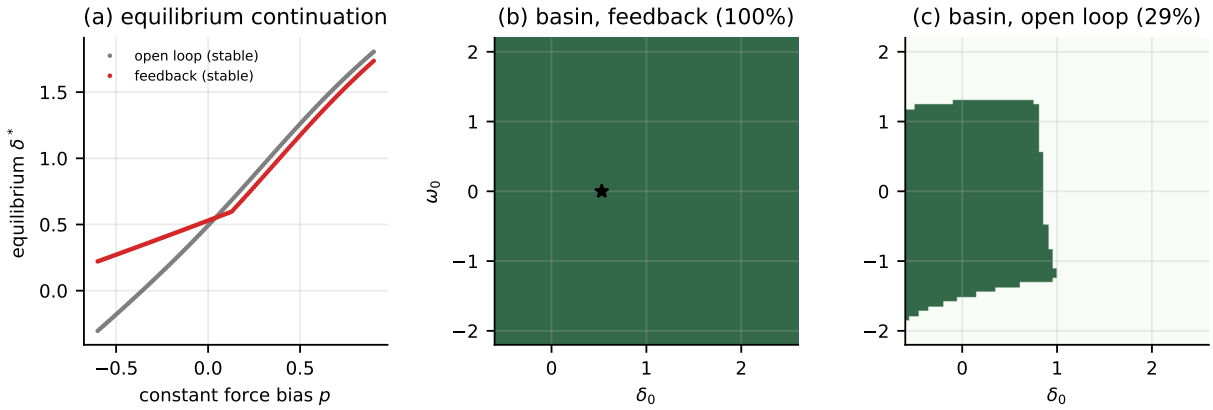


Figure S1: Scalar benchmark. (a) Equilibria versus constant force bias with stability from the finite-difference Jacobian. (b,c) Convergence to the nominal equilibrium from a 64×64 grid of initial conditions, with and without feedback.

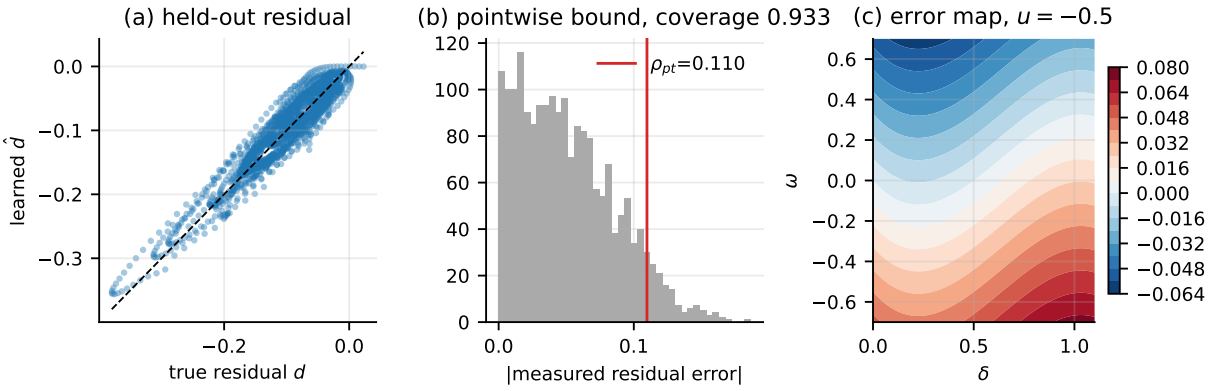


Figure S2: Scalar benchmark, residual identification. (a) Learned versus true residual on held-out points. (b) Distribution of the measured residual error with the pointwise bound ρ_{pt}. (c) Spatial error of the fit at $u = -0.5$.

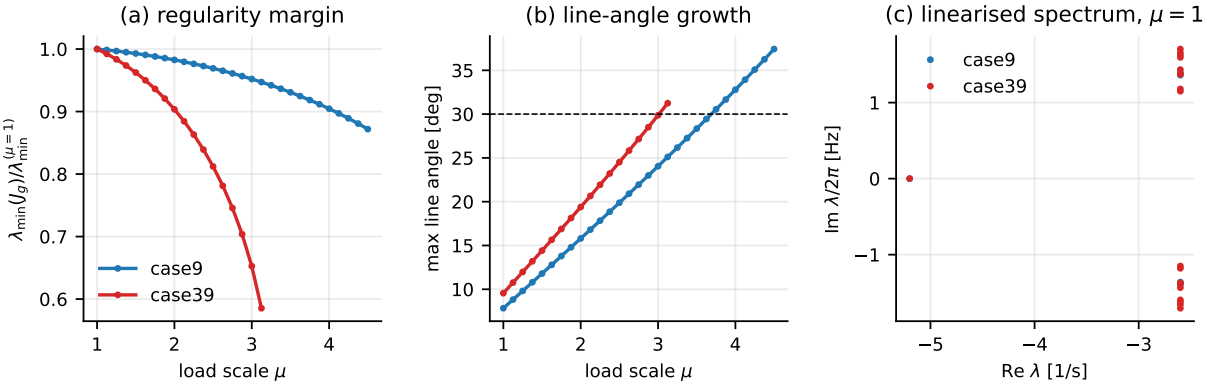


Figure S3: Network realizations. (a) Regularity margin along a uniform load-scale continuation. (b) Maximum line angle along the continuation; dashed: 30°. (c) Eigenvalues of the linearized plant at the nominal equilibrium (the neutral mode of the angle shift is omitted).

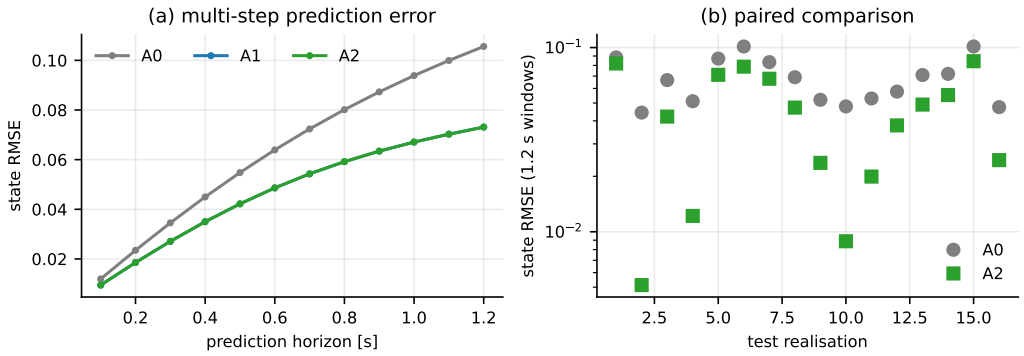


Figure S4: Scalar benchmark, prediction ablation on held-out closed-loop runs. (a) State RMSE versus prediction horizon for A0–A2. (b) Paired per-realization RMSE for A0 and A2 on a logarithmic axis.

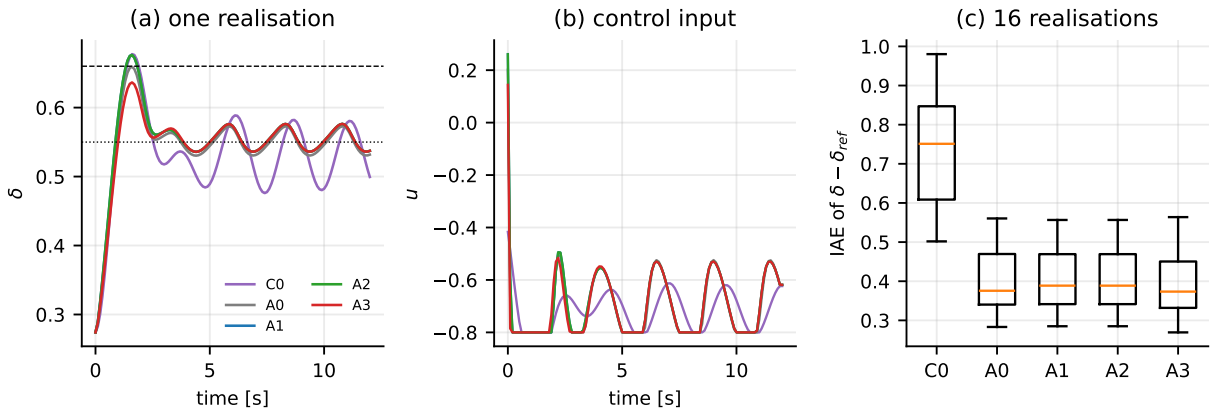


Figure S5: Scalar benchmark, closed loop. (a) One realization; dotted line: set point, dashed: angle limit. (b) Applied input. (c) IAE over 16 realizations.

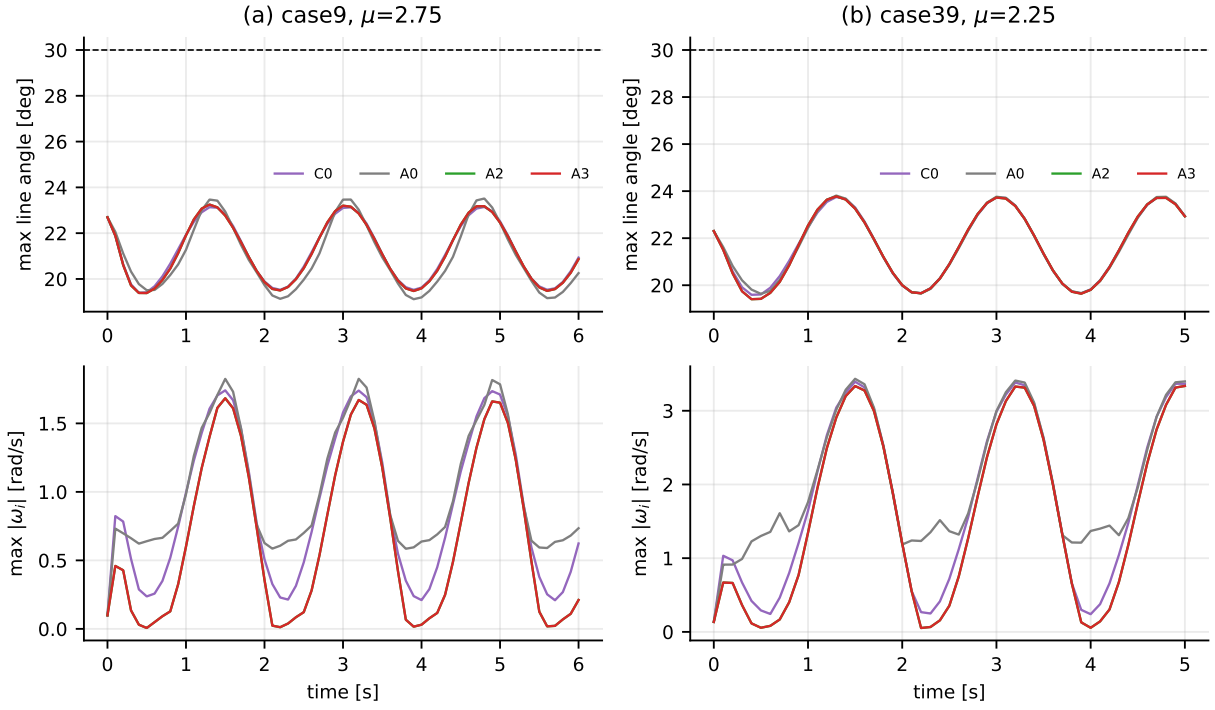


Figure S6: Network realizations, closed loop at elevated loading: (a) case9, (b) case39. Top: maximum line angle, with the 30° limit dashed. Bottom: maximum speed deviation. One realization per case; A2 and A3 coincide.

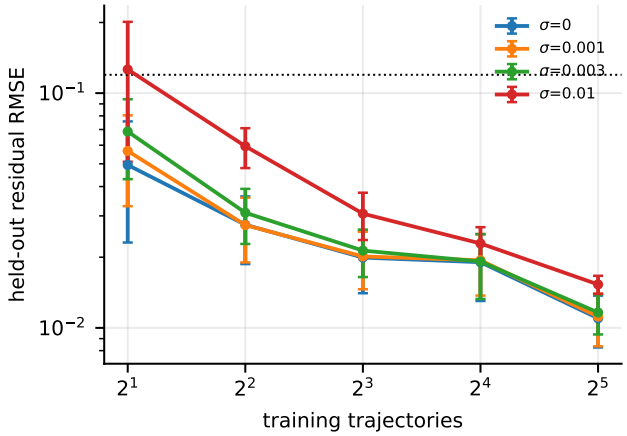


Figure S7: Held-out residual RMSE versus the number of training trajectories for several measurement-noise levels (mean ± s.d. over 4 draws); dotted: RMS of the residual itself.

Table 1: Closed loop after an unannounced step at 3 s: RMS speed deviation [rad/s] from 0.5 s after the step, and share of samples with the input at its bound. Six realizations for case9, four for case39. Physics + proj.: the fallback state alone ($\tau = 0$), run on case9 only. Oracle: A2 told the true dispatch.

		case9		case39	
Step μ	Controller	ω_{rms}	bound	ω_{rms}	bound
1.05	C0	0.344	0.00	0.767	0.00
	A0	0.178	0.22	0.541	0.60
	A2	0.073	0.19	0.458	1.00
	physics + proj.	0.178	0.22	–	–
	gate	0.017	0.08	0.463	0.81
	gate + reconc.	0.018	0.03	0.462	0.86
	oracle	0.015	0.03	0.462	0.85
1.10	C0	0.361	0.00	0.802	0.00
	A0	0.114	0.17	0.544	0.74
	A2	0.140	0.43	0.516	1.00
	physics + proj.	0.114	0.17	–	–
	gate	0.011	0.09	0.520	0.85
	gate + reconc.	0.023	0.06	0.524	0.90
	oracle	0.017	0.06	0.518	0.90

Table 2: Parameters of the network realizations and numerical settings. Items marked † are not in the MATPOWER files and are declared assumptions.

Quantity	Value
Generator inertia $M_i = 2HS_i/\omega_s$, $\omega_s = 2\pi \cdot 60$	$H = 4\,\mathrm{s}^{\dagger}$, $S_i = P_{\max,i}$
Generator–bus susceptance b_i	S_i/x'_d, $x'_d = 0.3$ p.u.†
Damping D_i (plant)	$4M_i$ s^{-1} †
Physics-model damping	$0.8D_i$
Plant-only effects	saliency $\kappa = 0.08$, friction $0.15D_i \tanh(2\omega_i)^{\dagger}$
Input bound	$\lvert u_i \rvert \le 0.1S_i^{\dagger}$
Dispatch	MATPOWER P_g scaled so that $\sum P_m = \sum P_d$
Sampling period T_s, plant step	0.1 s, 0.0125 s
Predictor sub-steps per T_s	2 (RK4)
Scalar benchmark plant step, sampling	0.01 s, 0.1 s
Network MPC horizon, iterations	$N = 5$, 12 L-BFGS-B iterations
Scalar MPC horizon, iterations	$N = 12$, 60 L-BFGS-B iterations
Conformal level	$\alpha = 0.05$
Bootstrap resamples	4000

1 Scalar benchmark

Table 3: Residual identification and conformal bound, scalar benchmark.

Quantity	Value
library_degree	3
library_terms	20
active_terms	3
ridge	0.0001
threshold	0.2
residual_rms_true_test	0.11971
residual_rms_error_after_fit	0.021155
error_reduction_percent	82.328
max_abs_error_test	0.066383
rho_pointwise_95	0.10974
coverage_pointwise_test	0.93281
rho_block_95	0.16133
coverage_block_test	0.95833
n_calibration_blocks	60
n_test_blocks	96

Table 4: Disturbance-amplitude sweep (8 realizations per amplitude). The advantage of A3 over A0 is visible only at small amplitudes.

Amplitude	IAE (A0)	Viol. (A0)	IAE (A3)	Viol. (A3)
0.00	0.368	0.0012	0.313	0.0000
0.04	0.371	0.0006	0.343	0.0000
0.08	0.403	0.0042	0.390	0.0005
0.12	0.485	0.0149	0.486	0.0089
0.22	0.899	0.1089	0.920	0.1083

Table 5: Learning curve of the scalar residual.

Noise s.d.	Training traj.	Held-out RMSE	s.d. (4 draws)
0	2	0.0494	0.0263
0	4	0.0275	0.0088
0	8	0.0199	0.0059
0	16	0.0191	0.0060
0	32	0.0110	0.0027
0.001	2	0.0567	0.0237
0.001	4	0.0274	0.0084
0.001	8	0.0201	0.0055
0.001	16	0.0194	0.0057
0.001	32	0.0112	0.0028
0.003	2	0.0686	0.0256
0.003	4	0.0309	0.0082
0.003	8	0.0213	0.0049
0.003	16	0.0192	0.0059
0.003	32	0.0116	0.0023
0.01	2	0.1261	0.0751
0.01	4	0.0594	0.0115
0.01	8	0.0307	0.0070
0.01	16	0.0229	0.0040
0.01	32	0.0153	0.0014

Table 6: Conformal envelope used by A3 in the scalar benchmark ($\alpha = 0.05$).

Step k	ε_k^{δ}	ε_k^{ω}
0	0.0000	0.0000
1	0.0012	0.0240
2	0.0048	0.0470
3	0.0105	0.0677
4	0.0182	0.0850
5	0.0274	0.0997
6	0.0375	0.1163
7	0.0482	0.1289
8	0.0609	0.1366
9	0.0747	0.1387
10	0.0885	0.1416
11	0.1015	0.1469
12	0.1131	0.1464

Table 7: Paired differences with respect to A0 (16 realizations, bootstrap 95% intervals).

Controller	Metric	Mean diff. vs A0	CI low	CI high
C0	iae	+0.3432	+0.2934	+0.3925
C0	last	+0.0249	+0.0193	+0.0312
C0	viol	+0.0255	+0.0109	+0.0420
A1	iae	-0.0012	-0.0094	+0.0060
A1	last	-0.0010	-0.0016	-0.0004
A1	viol	+0.0052	+0.0028	+0.0077
A2	iae	-0.0012	-0.0094	+0.0060
A2	last	-0.0010	-0.0016	-0.0004
A2	viol	+0.0052	+0.0028	+0.0077
A3	iae	-0.0071	-0.0181	+0.0067
A3	last	-0.0010	-0.0016	-0.0004
A3	viol	-0.0071	-0.0120	-0.0026

Table 8: Coverage of the scalar-benchmark envelope on independent trajectories (fraction of trajectories whose window-maximum error stays below the envelope).

Test source	Traj.	δ, k=6	ω, k=6	δ, k=12	ω, k=12	Joint, all k
static law	16	1.000	1.000	0.875	0.875	0.875
MPC (A2)	16	1.000	1.000	1.000	1.000	1.000

Table 9: Library complexity of the scalar residual (degree-3 library): held-out error against the number of active terms.

Threshold	Active terms	Held-out RMSE
0.002	20	0.0137
0.005	20	0.0137
0.01	20	0.0137
0.02	19	0.0137
0.05	18	0.0137
0.1	15	0.0136
0.2	4	0.0212
0.5	1	0.0216
1	1	0.0216

2 Network realizations

Table 10: Regularity and spectrum, case9.

Quantity	Value
buses	9
generators	3
equilibrium_residual	1.1102e-15
lambda_min_J_nominal	2.2374
cond_J_nominal	25.595
max_line_angle_deg_nominal	7.8349
largest_continued_load_scale	4.5
lambda_min_Lb_plus_G	2.25
cert_lower_bound_at_30deg	1.9486
slowest_damping_ratio	0.28457
max_real_part_nonneutral	-2.6
lowest_mode_freq_hz	1.3611
highest_mode_freq_hz	1.394

Table 11: Residual identification, case9.

Quantity	Nominal loading	Elevated loading
load_scale	1	2.75
train_traj	30	30
residual_rms_true_test	2.4036	6.3442
residual_rms_error_after_fit	0.057847	0.039624
error_reduction_percent	97.593	99.375
max_abs_error_test	1.8538	0.49743
rho_pointwise_95_measured	0.16014	0.16093
active_terms_total	24	34
terms_available	39	39

Table 12: One-factor sensitivity, case9. The plant parameter is scaled; predictors keep every nominal parameter, including the dispatch, and the nominal-loading residual.

Parameter	Scale	RMSE A0	RMSE A2	Ratio
load_scale	0.60	2.5763	1.5522	0.602
load_scale	0.80	1.6313	0.7422	0.455
load_scale	1.00	0.6940	0.1139	0.164
load_scale	1.25	0.5275	1.0357	1.963
load_scale	1.50	1.6949	2.8739	1.696
line_scale	0.60	0.6947	0.1184	0.170
line_scale	0.80	0.6942	0.1150	0.166
line_scale	1.00	0.6940	0.1139	0.164
line_scale	1.25	0.6939	0.1154	0.166
line_scale	1.50	0.6939	0.1200	0.173
inertia_scale	0.60	0.9157	0.2568	0.280
inertia_scale	0.80	0.7757	0.1425	0.184
inertia_scale	1.00	0.6940	0.1139	0.164
inertia_scale	1.25	0.6303	0.1318	0.209
inertia_scale	1.50	0.5887	0.1584	0.269
damping_scale	0.60	0.9129	0.2541	0.278
damping_scale	0.80	0.7750	0.1427	0.184
damping_scale	1.00	0.6940	0.1139	0.164
damping_scale	1.25	0.6304	0.1302	0.207
damping_scale	1.50	0.5887	0.1561	0.265

Table 13: Monte Carlo ensemble summary, case9.

Statistic	RMSE A0	RMSE A2	Ratio
count	24.0000	24.0000	24.0000
mean	0.4276	0.0727	0.1645
min	0.2658	0.0103	0.0302
max	0.6584	0.2209	0.3599

Table 14: Transfer across loadings, case9 (RMSE columns A0, A2 single, A2 pooled; load = load scale): the predictor knows loading and dispatch; the residual is fitted at nominal loading (single) or on five loadings (pooled).

Set	Load	A0	A2 single	A2 pooled	Ratio single	Ratio pooled
train	0.6	0.2478	0.0724	0.0733	0.292	0.296
train	1.0	0.4060	0.0668	0.0676	0.165	0.166
train	1.5	0.6328	0.2021	0.2036	0.319	0.322
interpolation	0.7	0.2853	0.0694	0.0701	0.243	0.246
interpolation	0.9	0.3574	0.0547	0.0555	0.153	0.155
interpolation	1.1	0.4351	0.0327	0.0331	0.075	0.076
interpolation	1.4	0.5607	0.0820	0.0828	0.146	0.148
extrapolation	1.8	0.7282	0.1576	0.1577	0.216	0.217

Table 15: Library complexity, case9: held-out residual error against the number of active terms (nominal loading).

Threshold	Active terms	Available	Held-out RMSE	Reduction [%]
0.002	39	39	0.0688	97.14
0.01	37	39	0.0688	97.14
0.03	33	39	0.0715	97.03
0.1	25	39	0.0579	97.59
0.3	18	39	0.0676	97.19
1	12	39	0.0698	97.10

Table 16: Plant mismatch term $0.15\, b_i \psi_i |\psi_i|$ outside the library, case9.

Quantity	Value
ool_coefficient	0.15
residual_rms_true_test	2.7136
residual_rms_error_after_fit	0.071091
error_reduction_percent	97.38
pred_ratio_A2_over_A0	0.17121
pred_ratio_lo	0.095234
pred_ratio_hi	0.25349
active_terms	29
mean_window_max_abs_g_A2	4.2286e-11

Table 17: Coverage of the line-angle envelope at the elevated loading, case9.

Test source	Traj.	Coverage, k=5	Joint, k=1 to 5
static law	4	1.00	1.00
MPC (A2)	3	1.00	1.00
closed-loop A2 realizations	6	1.00	1.00

Table 18: Trajectory-level envelope used by A3 at the elevated loading, case9.

Step k	ε_k [rad]
0	0.00000
1	0.01502
2	0.03133
3	0.04585
4	0.05849
5	0.06640

Table 19: Trajectory-level envelope of the largest line-angle difference at nominal loading, case9 (static-law calibration trajectories).

Step k	ε_k [rad]
0	0.00000
1	0.00262
2	0.00525
3	0.00729
4	0.00937
5	0.01032

Table 20: Regularity and spectrum, case39.

Quantity	Value
buses	39
generators	10
equilibrium_residual	6.2172e-15
lambda_min_J_nominal	4.1856
cond_J_nominal	246.57
max_line_angle_deg_nominal	9.5549
largest_continued_load_scale	3.125
lambda_min_Lb_plus_G	4.3023
cert_lower_bound_at_30deg	3.7259
slowest_damping_ratio	0.2359
max_real_part_nonneutral	-2.6
lowest_mode_freq_hz	1.1498
highest_mode_freq_hz	1.7046

Table 21: Residual identification, case39.

Quantity	Nominal loading	Elevated loading
load_scale	1	2.25
train_traj	30	30
residual_rms_true_test	5.4925	10.066
residual_rms_error_after_fit	0.088178	0.055325
error_reduction_percent	98.395	99.45
max_abs_error_test	3.0546	3.464
rho_pointwise_95_measured	0.19211	0.16093
active_terms_total	118	107
terms_available	130	130

Table 22: One-factor sensitivity, case39. The plant parameter is scaled; predictors keep every nominal parameter, including the dispatch, and the nominal-loading residual.

Parameter	Scale	RMSE A0	RMSE A2	Ratio
load_scale	0.60	5.7381	4.0659	0.709
load_scale	0.80	3.6645	1.8966	0.518
load_scale	1.00	1.6310	0.4890	0.300
load_scale	1.25	1.2728	2.5276	1.986
load_scale	1.50	3.8483	7.0805	1.840
line_scale	0.60	1.6320	0.4917	0.301
line_scale	0.80	1.6313	0.4897	0.300
line_scale	1.00	1.6310	0.4890	0.300
line_scale	1.25	1.6308	0.4900	0.300
line_scale	1.50	1.6307	0.4930	0.302
inertia_scale	0.60	2.1588	0.8157	0.378
inertia_scale	0.80	1.8290	0.5749	0.314
inertia_scale	1.00	1.6310	0.4890	0.300
inertia_scale	1.25	1.4742	0.4750	0.322
inertia_scale	1.50	1.3717	0.4956	0.361
damping_scale	0.60	2.1417	0.7898	0.369
damping_scale	0.80	1.8239	0.5676	0.311
damping_scale	1.00	1.6310	0.4890	0.300
damping_scale	1.25	1.4767	0.4766	0.323
damping_scale	1.50	1.3749	0.4963	0.361

Table 23: Monte Carlo ensemble summary, case39.

Statistic	RMSE A0	RMSE A2	Ratio
count	24.0000	24.0000	24.0000
mean	0.9379	0.1480	0.1593
min	0.5947	0.0054	0.0071
max	1.3459	0.3424	0.3033

Table 24: Transfer across loadings, case39 (RMSE columns A0, A2 single, A2 pooled; load = load scale): the predictor knows loading and dispatch; the residual is fitted at nominal loading (single) or on five loadings (pooled).

Set	Load	A0	A2 single	A2 pooled	Ratio single	Ratio pooled
train	0.6	0.5381	0.0908	0.0908	0.169	0.169
train	1.0	0.9084	0.2156	0.2157	0.237	0.237
train	1.5	1.2907	0.2223	0.2222	0.172	0.172
interpolation	0.7	0.6205	0.1248	0.1248	0.201	0.201
interpolation	0.9	0.8198	0.1711	0.1711	0.209	0.209
interpolation	1.1	0.9820	0.2060	0.2061	0.210	0.210
interpolation	1.4	1.2231	0.2962	0.2962	0.242	0.242
extrapolation	1.8	1.5209	0.4726	0.4715	0.311	0.310

Table 25: Library complexity, case39: held-out residual error against the number of active terms (nominal loading).

Threshold	Active terms	Available	Held-out RMSE	Reduction [%]
0.002	130	130	0.0832	98.49
0.01	130	130	0.0832	98.49
0.03	129	130	0.0832	98.48
0.1	124	130	0.0850	98.45
0.3	79	130	0.1042	98.10
1	50	130	0.1054	98.08

Table 26: Plant mismatch term $0.15\, b_i \psi_i |\psi_i|$ outside the library, case39.

Quantity	Value
ool_coefficient	0.15
residual_rms_true_test	7.1156
residual_rms_error_after_fit	0.073176
error_reduction_percent	98.972
pred_ratio_A2_over_A0	0.2166
pred_ratio_lo	0.15563
pred_ratio_hi	0.2775
active_terms	106
mean_window_max_abs_g_A2	4.0105e-11

Table 27: Coverage of the line-angle envelope at the elevated loading, case39.

Test source	Traj.	Coverage, k=5	Joint, k=1 to 5
static law	4	1.00	0.75
MPC (A2)	3	1.00	1.00
closed-loop A2 realizations	4	1.00	1.00

Table 28: Trajectory-level envelope used by A3 at the elevated loading, case39.

Step k	ε_k [rad]
0	0.00000
1	0.01565
2	0.02984
3	0.04161
4	0.05578
5	0.06362

Table 29: Trajectory-level envelope of the largest line-angle difference at nominal loading, case39 (static-law calibration trajectories).

Step k	ε_k [rad]
0	0.00000
1	0.00484
2	0.00946
3	0.01329
4	0.01665
5	0.01873

Table 30: Constraint-active test (case9, line-angle limit 23°).

Run	Ctrl.	Peak [deg]	Viol. integral	Frac. time over	ω_{rms}
0	C0	23.15	0.00122	0.098	1.002
0	A2	23.25	0.00202	0.115	0.871
0	A3	23.25	0.00203	0.115	0.871
1	C0	23.79	0.00180	0.066	1.069
1	A2	23.81	0.00214	0.098	0.942
1	A3	23.81	0.00215	0.098	0.942
2	C0	22.91	0.00000	0.000	0.983
2	A2	22.96	0.00000	0.000	0.826
2	A3	22.97	0.00000	0.000	0.826
3	C0	22.74	0.00000	0.000	1.044
3	A2	22.81	0.00000	0.000	0.888
3	A3	22.82	0.00000	0.000	0.888
4	C0	22.66	0.00000	0.000	1.025
4	A2	22.70	0.00000	0.000	0.862
4	A3	22.71	0.00000	0.000	0.862
5	C0	22.88	0.00000	0.000	1.036
5	A2	22.96	0.00000	0.000	0.866
5	A3	22.97	0.00000	0.000	0.866

3 Trust gate, reconciliation and recovery

The tables of this section are produced by the gate, reconciliation and atlas scripts of the repository and are read from its results folder. RMSE is the one-second state prediction error; the five predictors are A0, A2, A2 with the true dispatch (Oracle), the plain trust gate (Gate) and the gate with reconciliation (Reconc.).

Table 31: All prediction scenarios, case9. Step μ: scale of load and dispatch after the step at 3 s (persistent, or undone at 6 s for the recovery runs). $\hat{\rho}$: mean estimated dispatch scale; delay: time from the step until $q > 1$; $\tau_{\min}$ and re-trust: lowest trust and time from detection until $\tau \geq 0.8$.

Scenario	A0	A2	Oracle	Gate	Reconc.	$\hat{\rho}$	Delay [s]	$\tau_{\min}$	Re-trust [s]
no change	0.491	0.070	0.070	0.070	0.070	0.000	–	–	–
false-shift stress	0.504	0.124	0.124	0.124	0.124	0.000	–	–	–
persistent mu 1.03	0.425	0.179	0.159	0.175	0.161	1.030	0.28	0.96	0.10
persistent mu 1.05	0.350	0.175	0.116	0.143	0.127	1.050	0.18	0.89	0.10
persistent mu 1.1	0.221	0.300	0.131	0.169	0.141	1.104	0.10	0.59	0.13
persistent mu 0.8	1.166	0.543	0.134	1.046	0.138	0.799	0.10	0.30	0.50
persistent mu 1.25	0.370	0.725	0.101	0.407	0.108	1.251	0.10	0.20	0.50
persistent mu 1.5	1.200	1.992	0.107	1.203	0.111	1.499	0.10	0.03	0.50
recovery mu 1.1	0.199	0.295	0.114	0.145	0.115	1.101	0.10	0.61	0.10
recovery mu 1.25	0.418	0.756	0.182	0.452	0.202	1.250	0.10	0.21	0.50

Table 32: Recovery runs, case9: prediction RMSE from 0.6 s after the step is undone.

Scenario	A0	A2	Oracle	Gate	Reconc.
recovery mu 1.1	0.509	0.098	0.098	0.098	0.097
recovery mu 1.25	0.524	0.136	0.136	0.136	0.140

Table 33: Large unannounced or announced step to $\mu = 1.5$ at 3 s in closed loop, case9 (six realizations for case9, four for case39): RMS speed deviation from 0.5 s after the step [rad/s], input energy, and mean time per control step [ms]. With the dispatch unannounced the input is at its bound in all predictive controllers and the variants cannot be told apart.

Dispatch	Controller	ω_{rms}	E_u	Time [ms]
unannounced	C0	0.514	1.5	0.0
unannounced	A0	0.260	7.8	15.6
unannounced	A2	0.255	7.8	7.4
unannounced	physics + projection	0.260	7.8	7.5
unannounced	gate	0.255	7.8	7.5
announced	C0	0.514	1.5	0.0
announced	A0	0.391	4.3	25.3
announced	A2	0.139	5.6	10.0
announced	physics + projection	0.391	4.3	13.4
announced	gate	0.139	5.6	9.4

Table 34: Validation score of the gate, case9: mean ratio of the gated to the A0 error over the validation scenarios, for the innovation constant a (rows) and the weight b of the regularity score that was tried and not kept (columns).

a	$b = 0$	$b = 0.5$	$b = 1$	$b = 2$
0.25	0.6687	0.6686	0.6686	0.6684
0.5	0.6695	0.6694	0.6694	0.6693
1	0.6711	0.6710	0.6710	0.6709
2	0.6711	0.6711	0.6710	0.6710

Table 35: All prediction scenarios, case39. Step μ: scale of load and dispatch after the step at 3 s (persistent, or undone at 6 s for the recovery runs). $\hat{\rho}$: mean estimated dispatch scale; delay: time from the step until $q > 1$; $\tau_{\min}$ and re-trust: lowest trust and time from detection until $\tau \geq 0.8$.

Scenario	A0	A2	Oracle	Gate	Reconc.	$\hat{\rho}$	Delay [s]	$\tau_{\min}$	Re-trust [s]
no change	1.082	0.101	0.101	0.101	0.101	0.000	–	–	–
false-shift stress	1.183	0.431	0.431	0.432	0.458	0.005	–	–	–
persistent mu 1.03	0.953	0.358	0.321	0.360	0.341	1.032	0.23	0.91	0.10
persistent mu 1.05	0.800	0.440	0.316	0.353	0.317	1.049	0.10	0.73	0.10
persistent mu 1.1	0.436	0.713	0.268	0.285	0.269	1.101	0.10	0.43	0.50
persistent mu 0.8	2.568	1.293	0.143	2.495	0.144	0.800	0.10	0.15	0.50
persistent mu 1.25	0.992	1.893	0.453	1.011	0.479	1.251	0.10	0.08	0.50
persistent mu 1.5	2.670	4.998	0.310	2.671	0.321	1.504	0.10	0.00	0.50
recovery mu 1.1	0.478	0.574	0.197	0.227	0.203	1.100	0.10	0.43	0.50
recovery mu 1.25	0.921	1.859	0.265	0.944	0.284	1.250	0.10	0.08	0.50

Table 36: Recovery runs, case39: prediction RMSE from 0.6 s after the step is undone.

Scenario	A0	A2	Oracle	Gate	Reconc.
recovery mu 1.1	1.072	0.150	0.150	0.150	0.152
recovery mu 1.25	1.113	0.222	0.222	0.222	0.236

Table 37: Large unannounced or announced step to $\mu = 1.5$ at 3 s in closed loop, case39 (six realizations for case9, four for case39): RMS speed deviation from 0.5 s after the step [rad/s], input energy, and mean time per control step [ms]. With the dispatch unannounced the input is at its bound in all predictive controllers and the variants cannot be told apart.

Dispatch	Controller	ω_{rms}	E_u	Time [ms]
unannounced	C0	1.149	14.7	0.0
unannounced	A0	1.018	21.0	292.8
unannounced	A2	1.017	21.0	19.2
unannounced	physics + projection	1.018	21.0	51.2
unannounced	gate	1.017	21.0	19.6
announced	C0	1.149	14.7	0.0
announced	A0	1.127	16.0	471.4
announced	A2	1.018	20.9	20.1
announced	physics + projection	1.127	16.0	79.0
announced	gate	1.018	20.9	19.1

Table 38: Validation score of the gate, case39: mean ratio of the gated to the A0 error over the validation scenarios, for the innovation constant a (rows) and the weight b of the regularity score that was tried and not kept (columns).

a	$b = 0$	$b = 0.5$	$b = 1$	$b = 2$
0.25	0.7080	0.7073	0.7068	0.7062
0.5	0.7077	0.7071	0.7066	0.7060
1	0.7078	0.7072	0.7067	0.7062
2	0.7078	0.7072	0.7067	0.7062

4 Nuisance-shift test of the trust gate

The plant has the stated parameter error for the whole run; the predictor and the gate keep the nominal parameters. Noise 10^{-3} on every state; case9 six runs, case39 four runs. Fallback: share of samples with $\tau < 0.5$; $\max|\rho-1|$: largest deviation of the estimated dispatch scale; ratio: A2 over A0 prediction error.

Table 39: Nuisance-shift test, case9.

Scenario	Fallback [%]	Mean τ	$\max\|\rho-1\|$	Ratio
line scale 0.85	0.0	0.869	0.021	0.23
line scale 1.15	0.0	0.864	0.021	0.29
inertia scale 0.8	1.0	0.958	0.077	0.22
inertia scale 1.2	0.0	0.994	0.039	0.23
damping scale 0.8	0.0	0.988	0.024	0.16
damping scale 1.2	0.0	0.997	0.028	0.24
joint (lines 1.12, inertia 0.85, damping 1.15)	0.0	0.931	0.035	0.22

Table 40: Nuisance-shift test, case39.

Scenario	Fallback [%]	Mean τ	$\max\|\rho-1\|$	Ratio
line scale 0.85	0.0	0.897	0.013	0.22
line scale 1.15	0.0	0.884	0.018	0.20
inertia scale 0.8	0.6	0.888	0.061	0.28
inertia scale 1.2	0.0	0.976	0.035	0.26
damping scale 0.8	0.0	0.924	0.029	0.21
damping scale 1.2	0.0	0.952	0.025	0.32
joint (lines 1.12, inertia 0.85, damping 1.15)	0.0	0.948	0.020	0.23

5 Held-out dispatch family

The load steps by the factor μ at 3 s, and after the step each generator's dispatch is the scaled dispatch times a random factor in $[0.8, 1.2]$, renormalized to keep the power balance. One-second prediction RMSE from 0.6 s after the step for A0, A2, A2 with the true dispatch (Oracle), the plain gate and the reconciling gate; $\hat{\rho}$: mean estimated common dispatch scale; all steps were detected within 0.1 s.

Table 41: Held-out family with a non-uniform dispatch, case9 (six runs for case9, four for case39).

Step μ	A0	A2	Oracle	Gate	Reconc.	$\hat{\rho}$
1.1	0.197	0.294	0.108	0.129	0.187	1.094
1.25	0.381	0.727	0.144	0.413	0.222	1.252

Table 42: Held-out family with a non-uniform dispatch, case39 (six runs for case9, four for case39).

Step μ	A0	A2	Oracle	Gate	Reconc.	$\hat{\rho}$
1.1	0.350	0.670	0.113	0.205	0.796	1.104
1.25	0.841	1.781	0.209	0.861	0.940	1.245

6 Reproduction

The repository contains `code/`, `data/` (the MATPOWER files), `results/`, `figures/` and `tests/`. With Python ≥ 3.10 and the packages in `requirements.txt`, `bash run_all.sh` runs the consistency tests, the scalar experiments, the network experiments and the constraint-active test, and regenerates every CSV and figure (the scripts of the exact gradient, the validity atlas, the trust gate and the reconciliation are included; the gate and recovery experiments take about one hour on one core). All random numbers use fixed seeds written in the scripts; no external data are downloaded.

7 Software environment

Python 3, NumPy, SciPy (L-BFGS-B, discrete algebraic Riccati and Lyapunov solvers, a Levenberg–Marquardt root finder; all time integrators are explicit RK4 written in the code), Matplotlib, pandas and Numba (optional; without Numba the code runs unchanged but much more slowly).